CAPSTONE PROJECT (1)

Health Care Project_Insurance Cost

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1. Problem Understanding

a) Defining problem statement b) Need of the study/project c) Understanding business/social opportunity

a) Defining the Problem Statement:
Healthcare costs are rising, making it essential for people to have insurance to cover these expenses. At the same time, insurance companies need to manage their financial risk by setting premiums that reflect each person's health and lifestyle choices. This will help insurance companies set fair prices and ensure that both individuals and insurers are financially protected.
4o
b) Need for the Study/Project:
Medical treatment can be very expensive, especially for those without insurance. A fair and accurate model to estimate insurance costs would benefit both individuals and insurance companies. It allows individuals to get a premium based on their specific health and lifestyle, and helps insurers manage risks by considering factors like medical history and habits. This not only protects people from unexpected costs but also encourages healthier choices by offering lower premiums for healthier lifestyles.
c) Understanding Business/Social Opportunity:
This project meets a key need in healthcare by helping insurers set fair and affordable insurance costs. A data-driven model for pricing can encourage healthier lifestyles by offering lower premiums to healthier individuals. For insurers, it also supports a sustainable business by helping them better understand customer health risks, manage costs, and retain customers. This benefits society by making healthcare coverage more accessible and promoting healthier habits.

2. Data Report

a) Understanding how data was collected in terms of time, frequency and methodology b) Visual inspection of data (rows, columns, descriptive details) c) Understanding of attributes (variable info, renaming if required)

a) Understanding How Data Was Collected:
The dataset, includes health, lifestyle, and demographic information for people applying for insurance. This data was likely gathered during the application process, with details on health checks, doctor visits, and metrics like BMI and cholesterol. Data collection likely happens yearly, as it covers information from the past year for each individual.
Target Variables: Insurance_Cost
Data dictionary:

Variable	Business Definition
applicant_id	Applicant unique ID
years_of_insurance_with_us	Since how many years customer is taking policy from the same company only
regular_checkup_lasy_year	Number of times customers has done the regular health check up in last one year
adventure_sports	Customer is involved with adventure sports like climbing, diving etc.
Occupation	Occupation of the customer
visited_doctor_last_1_year	Number of times customer has visited doctor in last one year
cholesterol_level	Cholesterol level of the customers while applying for insurance
daily_avg_steps	Average daily steps walked by customers
age	Age of the customer
heart_decs_history	Any past heart diseases
other_major_decs_history	Any past major diseases apart from heart like any operation
Gender	Gender of the customer
avg_glucose_level	Average glucose level of the customer while applying the insurance
bmi	BMI of the customer while applying the insurance
smoking_status	Smoking status of the customer
Year_last_admitted	When customer have been admitted in the hospital last time
Location	Location of the hospital
weight	Weight of the customer
covered_by_any_other_company	Customer is covered from any other insurance company
Alcohol	Alcohol consumption status of the customer
exercise	Regular exercise status of the customer
weight_change_in_last_one_year	How much variation has been seen in the weight of the customer in last year
fat_percentage	Fat percentage of the customer while applying the insurance
insurance_cost	Total Insurance cost

Table1- Data Dictionary

b) Visual inspection of data (rows, columns, descriptive details):

- Top 5 Rows of the dataset

	applicant_id	years_of_insurance_with_us	regular_checkup_lasy_year	adventure_sports	Occupation	visited_doctor_last_1_year	cholesterol_level	daily_avg_steps	age	h
0	5000	3	1	1	Salried	2	125 to 150	4866	28	
1	5001	0	0	0	Student	4	150 to 175	6411	50	
2	5002	1	0	0	Business	4	200 to 225	4509	68	
3	5003	7	4	0	Business	2	175 to 200	6214	51	
4	5004	3	1	0	Student	2	150 to 175	4938	44	

moking_status	Year_last_admitted	Location	weight	covered_by_any_other_company	Alcohol	exercise	weight_change_in_last_one_year	fat_percentage	insurance_cost
Unknown	NaN	Chennai	67	N	Rare	Moderate	1	25	20978
formerly smoked	NaN	Jaipur	58	N	Rare	Moderate	3	27	6170
formerly smoked	NaN	Jaipur	73	N	Daily	Extreme	0	32	28382
Unknown	NaN	Chennai	71	Y	Rare	No	3	37	27148
never smoked	2004.0	Bangalore	74	N	No	Extreme	0	34	29616

Table2- Top 5 rows of the dataset

- Last 5 Rows of the dataset

	applicant_id	years_of_insurance_with_us	regular_checkup_lasy_year	adventure_sports	Occupation	visited_doctor_last_1_year	cholesterol_level	daily_avg_steps	ag
24995	29995	3	0	0	Salried	4	225 to 250	5614	2
24996	29996	6	0	0	Business	4	200 to 225	4719	5
24997	29997	7	0	1	Student	2	150 to 175	5624	3
24998	29998	1	0	0	Salried	2	225 to 250	10777	2
24999	29999	8	2	0	Business	4	150 to 175	5882	2

oking_status	Year_last_admitted	Location	weight	covered_by_any_other_company	Alcohol	exercise	weight_change_in_last_one_year	fat_percentage	insurance_cost
smokes	2000.0	Kanpur	79	Y	Rare	Moderate	4	40	39488
ever smoked	2009.0	Kanpur	66	N	Rare	Moderate	2	28	14808
Unknown	NaN	Bhubaneswar	76	N	Rare	Moderate	1	35	33318
Unknown	NaN	Surat	74	N	Rare	Moderate	0	40	30850
formerly smoked	2014.0	Chennai	57	N	No	No	4	21	6170

Table3- Last 5 rows of the

>Observations; The dataset contained 25000 rows and 24 columns.

- **Basic Info of the dataset**

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25000 entries, 0 to 24999
Data columns (total 24 columns):
 #   Column                          Non-Null Count  Dtype
---  ------                          --------------  -----
 0   applicant_id                    25000 non-null  int64
 1   years_of_insurance_with_us      25000 non-null  int64
 2   regular_checkup_lasy_year       25000 non-null  int64
 3   adventure_sports                25000 non-null  int64
 4   Occupation                      25000 non-null  object
 5   visited_doctor_last_1_year      25000 non-null  int64
 6   cholesterol_level               25000 non-null  object
 7   daily_avg_steps                 25000 non-null  int64
 8   age                             25000 non-null  int64
 9   heart_decs_history              25000 non-null  int64
 10  other_major_decs_history        25000 non-null  int64
 11  Gender                          25000 non-null  object
 12  avg_glucose_level               25000 non-null  int64
 13  bmi                             24010 non-null  float64
 14  smoking_status                  25000 non-null  object
 15  Year_last_admitted              13119 non-null  float64
 16  Location                        25000 non-null  object
 17  weight                          25000 non-null  int64
 18  covered_by_any_other_company    25000 non-null  object
 19  Alcohol                         25000 non-null  object
 20  exercise                        25000 non-null  object
 21  weight_change_in_last_one_year  25000 non-null  int64
 22  fat_percentage                  25000 non-null  int64
 23  insurance_cost                  25000 non-null  int64
dtypes: float64(2), int64(14), object(8)
memory usage: 4.6+ MB
```

Table4- Basic Info of the dataset

>Observations; In this dataset there are 2 float64, 14 int64 and 8 object datatypes are present. BMI and Year_last_admitted shows missing values are present in those columns.

- **Checking Missing values**

```
applicant_id                        0
years_of_insurance_with_us          0
regular_checkup_lasy_year           0
adventure_sports                    0
Occupation                          0
visited_doctor_last_1_year          0
cholesterol_level                   0
daily_avg_steps                     0
age                                 0
heart_decs_history                  0
other_major_decs_history            0
Gender                              0
avg_glucose_level                   0
bmi                               990
smoking_status                      0
Year_last_admitted              11881
Location                            0
weight                              0
covered_by_any_other_company        0
Alcohol                             0
exercise                            0
weight_change_in_last_one_year      0
fat_percentage                      0
insurance_cost                      0
dtype: int64
```

Table5- Missing value of the dataset

>Observations; 'bmi' column have present 990 missing values and 'Year_last_admitted' column have present 11881 missing values. We see that the 'applicant_id' and 'Year_last_admitted' are not going to relevant for the analysis so we dropped these variables. We treat the missing values of the 'bmi' column with its mean value.

- Statistical Summary of the dataset

	count	mean	std	min	25%	50%	75%	max
applicant_id	25000.0	17499.500000	7217.022701	5000.0	11249.75	17499.5	23749.25	29999.0
years_of_insurance_with_us	25000.0	4.089040	2.606612	0.0	2.00	4.0	6.00	8.0
regular_checkup_lasy_year	25000.0	0.773680	1.199449	0.0	0.00	0.0	1.00	5.0
adventure_sports	25000.0	0.081720	0.273943	0.0	0.00	0.0	0.00	1.0
visited_doctor_last_1_year	25000.0	3.104200	1.141663	0.0	2.00	3.0	4.00	12.0
daily_avg_steps	25000.0	5215.889320	1053.179748	2034.0	4543.00	5089.0	5730.00	11255.0
age	25000.0	44.918320	16.107492	16.0	31.00	45.0	59.00	74.0
heart_decs_history	25000.0	0.054640	0.227281	0.0	0.00	0.0	0.00	1.0
other_major_decs_history	25000.0	0.098160	0.297537	0.0	0.00	0.0	0.00	1.0
avg_glucose_level	25000.0	167.530000	62.729712	57.0	113.00	168.0	222.00	277.0
bmi	24010.0	31.393328	7.876535	12.3	26.10	30.5	35.60	100.6
Year_last_admitted	13119.0	2003.892217	7.581521	1990.0	1997.00	2004.0	2010.00	2018.0
weight	25000.0	71.610480	9.325183	52.0	64.00	72.0	78.00	96.0
weight_change_in_last_one_year	25000.0	2.517960	1.690335	0.0	1.00	3.0	4.00	6.0
fat_percentage	25000.0	28.812280	8.632382	11.0	21.00	31.0	36.00	42.0
insurance_cost	25000.0	27147.407680	14323.691832	2468.0	16042.00	27148.0	37020.00	67870.0

Table6- Statistical Summary of the dataset

>Observations;
* Applicants: 25,000 people, with IDs from 5,000 to 29,999.
* Insurance Duration: Average of 4 years, ranging from 0 to 8 years.
* Health Checkups: 77% had a checkup last year, and only 8% do adventure sports.
* Doctor Visits: About 3 visits per year on average, with some up to 12.
* Physical Activity: Average daily steps are 5,216, ranging from 2,034 to 11,255.
* Age: Average age is 45, with a range from 16 to 74 years.
* Health Conditions: 5% have heart disease; 10% have other major illnesses.
* Glucose & BMI: Average glucose level is 168 mg/dL, and BMI is around 31 (high).
* Hospital Admissions: For those with admissions, the last was around 2003 on average.
* Weight: Average weight is 72 kg, ranging mostly between 64 and 78 kg.
* Weight Change: Average change is 2.5 kg in the last year.
* Fat Percentage: Average body fat is 29%, with a range from 11% to 42%.
* Insurance Cost: Average cost is $27,147, with a range from $2,468 to $67,870.

c) Understanding of attributes

Each attribute provides important information for assessing health risks:

- Years of Insurance: Shows loyalty or stability with the insurer.
- Regular Checkup and Doctor Visits: Indicate proactive health management.
- Adventure Sports: Higher-risk activities that might increase insurance cost.
- Occupation: Some jobs may impact lifestyle and health.
- Health Metrics (cholesterol, glucose, BMI): Direct indicators of health status.
- Medical History (heart disease, major surgeries): Key predictors of health risk.
- Lifestyle Factors (smoking, alcohol, exercise): Strongly linked to health outcomes.
- Weight, Weight Change, Fat Percentage: Help assess potential health risks.
- Insurance Cost: The target variable, showing the insurance cost for each person

3. Exploratory Data Analysis

a) Univariate Analysis

- Boxplots for all numerical columns

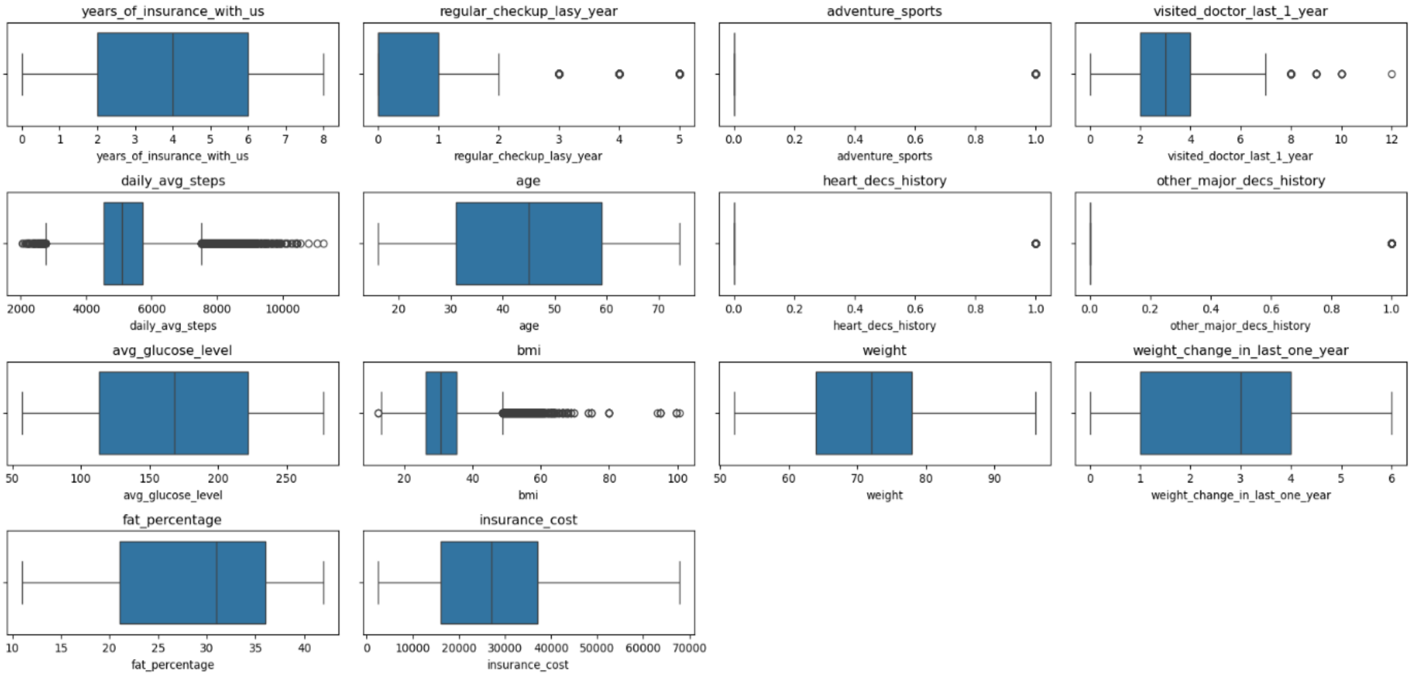

Figure1- Boxplots for all numerical variables

>Observations;

* In the column years_of_insurance_with_us most people have lower years with insurance and a few have high values.
* In the column regular_checkup_last_year most people had fewer checkups, with a few having more than average.
* In the column adventure_sports this looks like a yes-or-no (0 or 1) variable.
* In the column visited_doctor_last_year most people visited the doctor a few times, but some visited much more often.
* In the column daily_avg_steps most people have an average of 4000-8000 steps daily, but some have very low or very high values.
* In the column age are spread out evenly, with no major outliers.
* In the column heart_decs_history and other_major_decs_history these seem to be yes-or-no variables for health conditions.
* In the column avg_glucose_level most people have a lower glucose level, with a few high values.
* In the column bmi has a few high outliers, but most values are below that range.
* In the column Weight are spread fairly evenly with no major outliers.
* In the column weight_change_in_last_one_year weight changes are spread evenly with no major outliers.
* In the column fat_percentage vary widely with no big outliers.
* In the column insurance_cost vary a lot, with no significant outliers.

- Distributions of all Continuous Columns

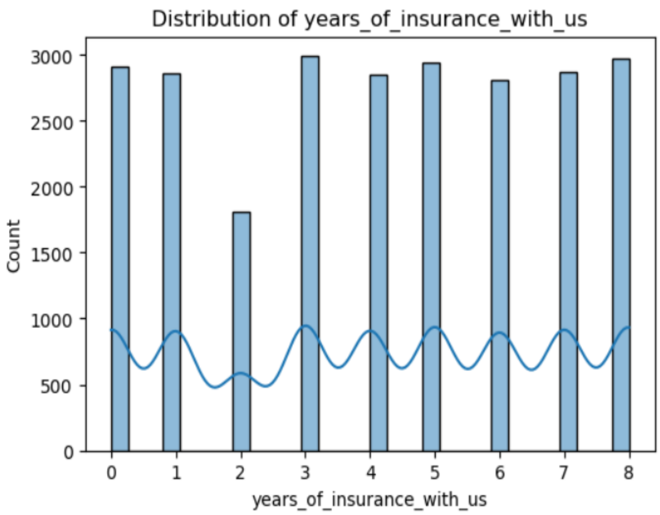

Figure2- Distributions of years of insurance with us

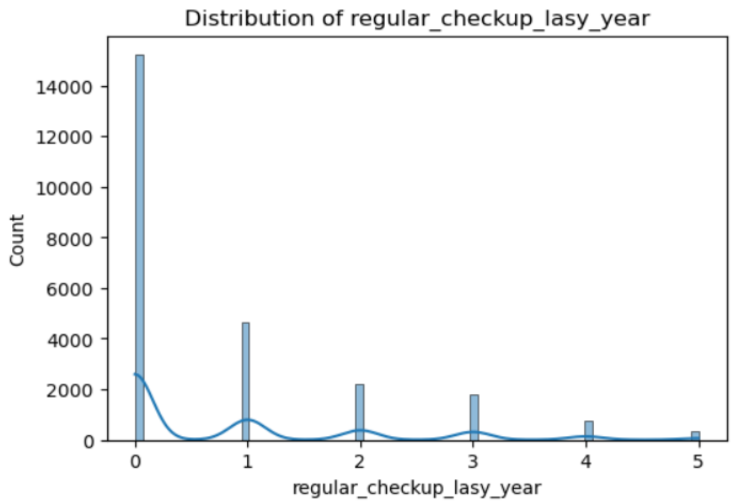

Figure3- Distribution of regular checkup lasy year

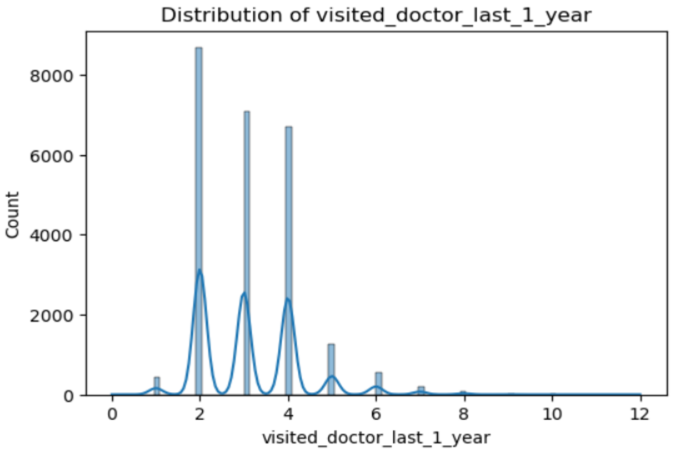

Figure4- Distribution of visited doctor last 1 year

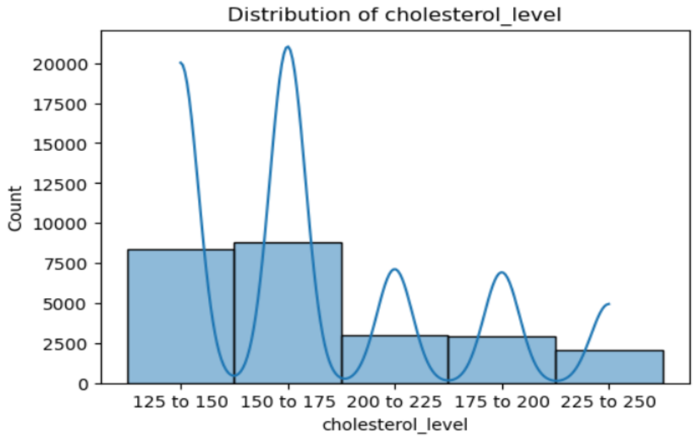

Figure5- Distribution of cholesterol level

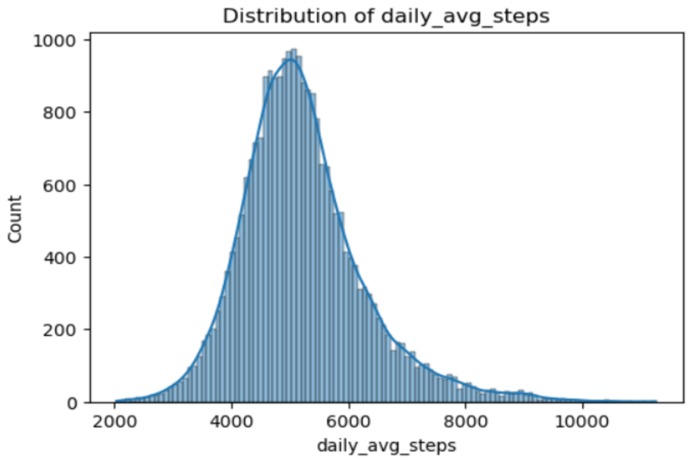

Figure6- Distributions of daily avg steps

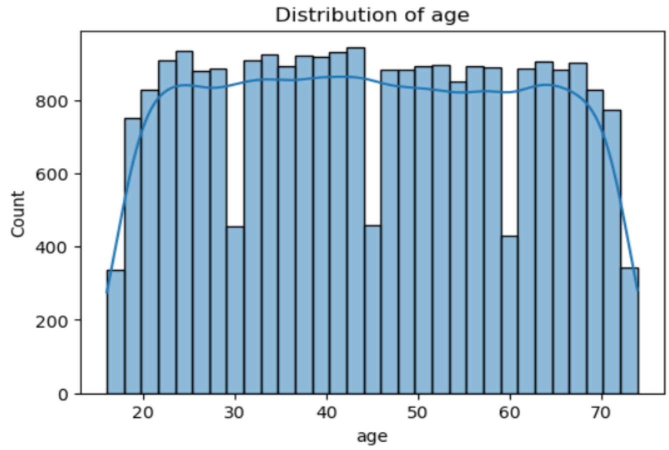

Figure7- Distribution of Age

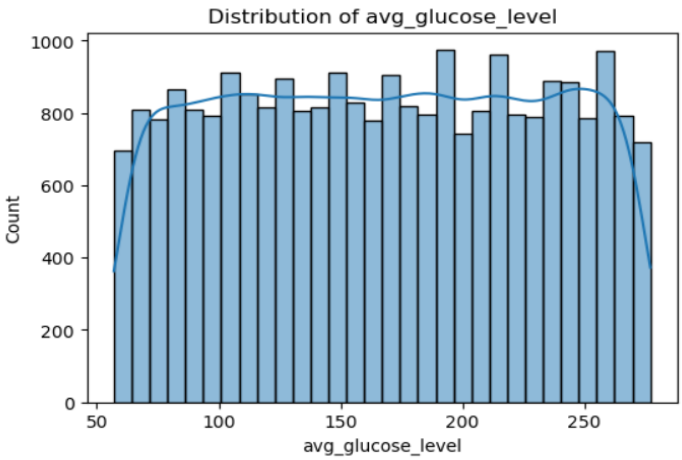

Figure8- Distribution of avg glucose level

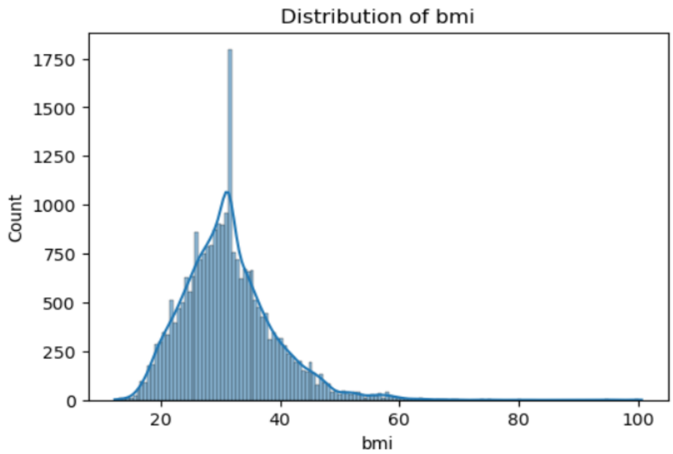

Figure9- Distribution of BMI

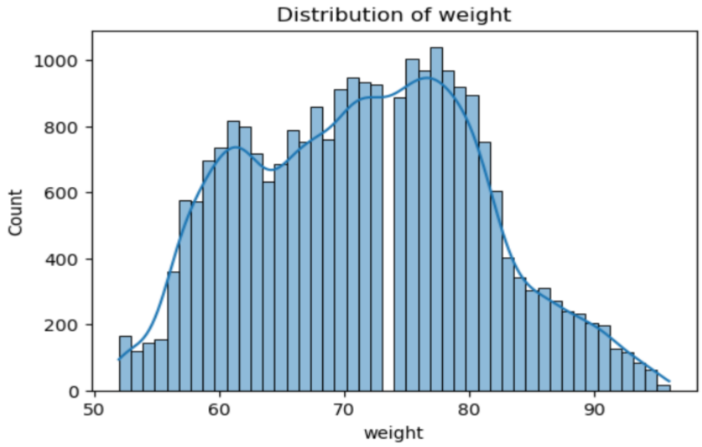

Figure10- Distribution of Weight

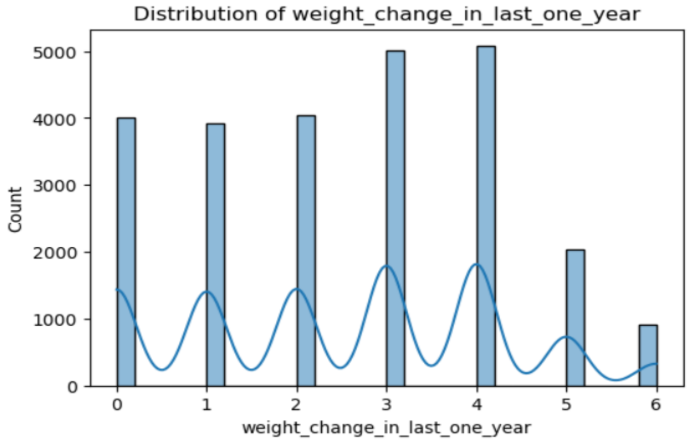

Figure11- Distribution of weight change in last one year

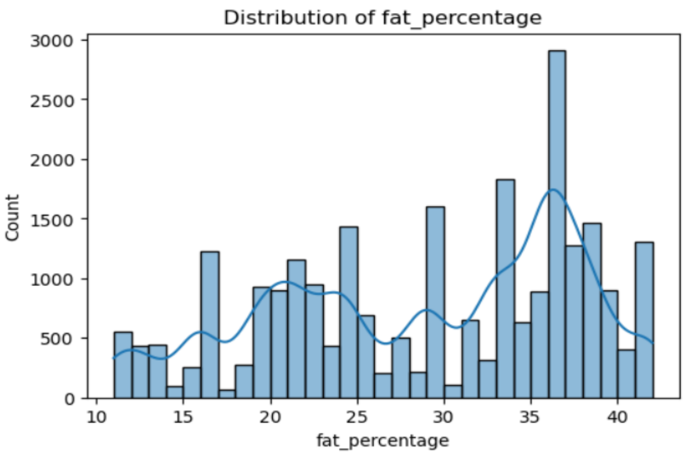

Figure12- Distribution of fat percentage

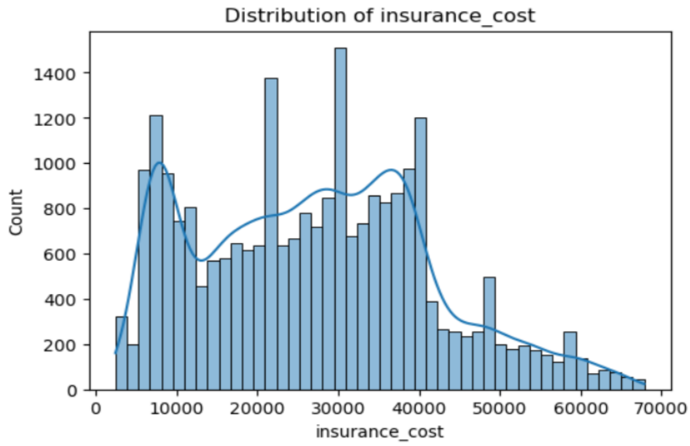

Figure13- Distribution of insurance cost

>Observations;

* We see in the column Years of Insurance with Us:
  - For the right-skewed we can say most customers are new.
  - For the left-skewed we can say many customers have been with us long-term.
* We see in the column Regular Checkup Lasy Year:
  - Peaks of the bins around 0 or 1 suggested that people either consistently had checkups or didn't.
  - Balanced of the bins suggested that a mix of people who did and didn't get checkups.
* We see in the column Visited Doctor Last 1 Year:
  - Fos the right-skewed we can say most visited the doctor often.
  - For the left-skewed we can say few visited the doctor.
* We see in the column Cholesterol Level:
  - Normal bins shows healthy cholesterol levels.
  - For the right-skewed we can say many have high cholesterol.
* We see in the column Daily Average Steps:
  - Peak at low values: Many people are inactive.
  - Right-skewed or balanced: Higher levels of physical activity.
* We see in the column Age:
  - For the right-skewed we can say mostly younger people.
  - For the left-skewed we can say mostly older people.
* We see in the column Average Glucose Level:
  - Normal distribution of the bis suggested a healthy glucose levels.
  - Skewed bins suggested that the glucose levels vary by health status.
* We see in the column BMI:
  - For the right-skewed we can say many people are overweight.
* We see in the column Weight:
  - Likely to match BMI trends, depending on population.
* We see in the column Weight Change in Last Year:
  - Peaks of the bins around 0 indicate stable weight.
  - For the help of positive/negative values we can understood people gained or lost weight.
* We see in the column Fat Percentage:

- For the right-skewed we can say many people have higher fat percentages.

* We see in the column Insurance Cost:

- For the right-skewed we can see that a few high-cost cases drive up average cost.

- Distributions of all Categorical Columns

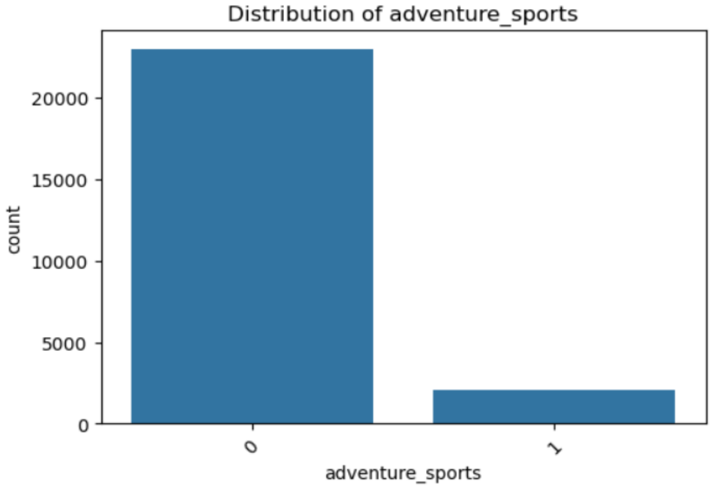

Figure14- distribution of adventure sports

Figure15- Distribution of Occupation

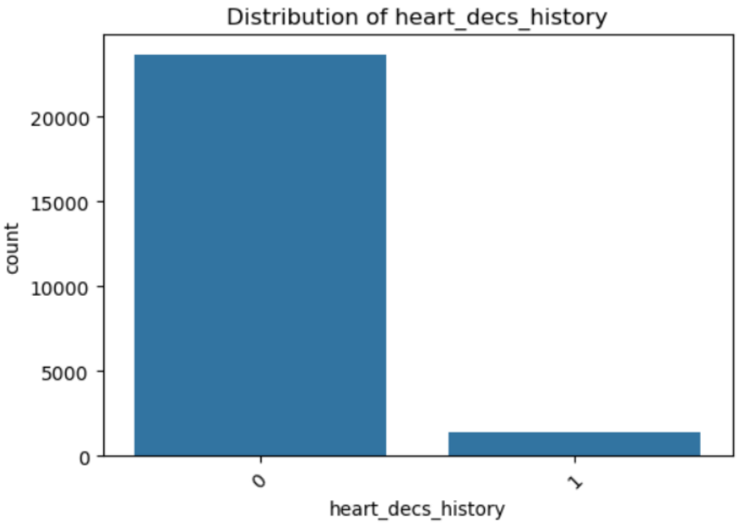

Figure16- Distribution of heart decs history

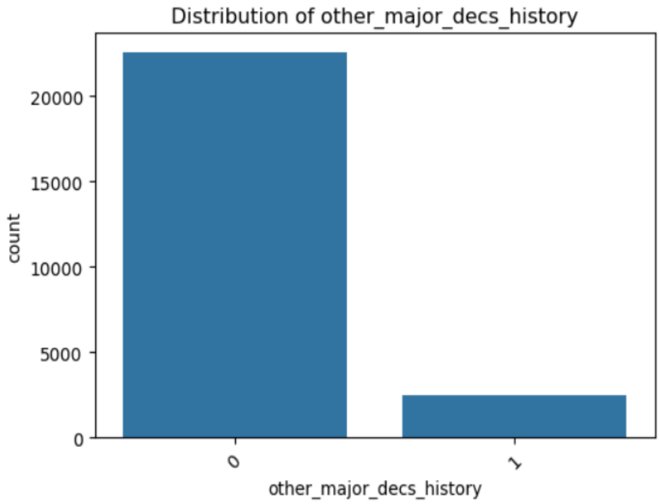

Figure17- Distribution of other major decs history

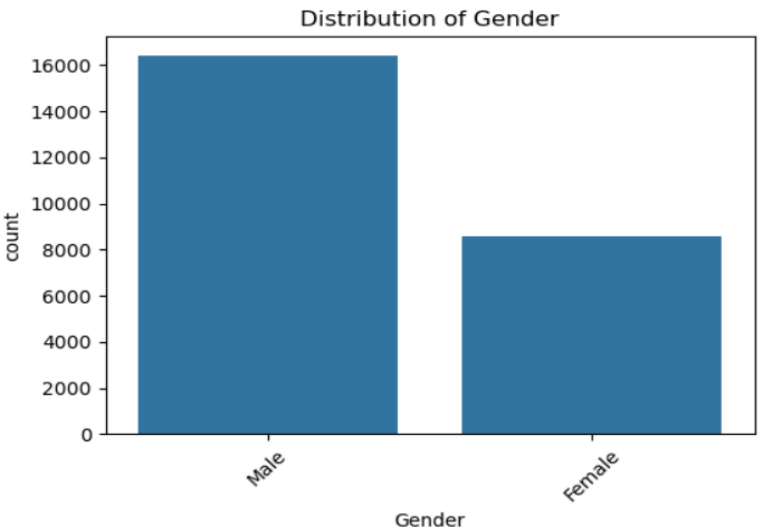

Figure18- Distribution of gender

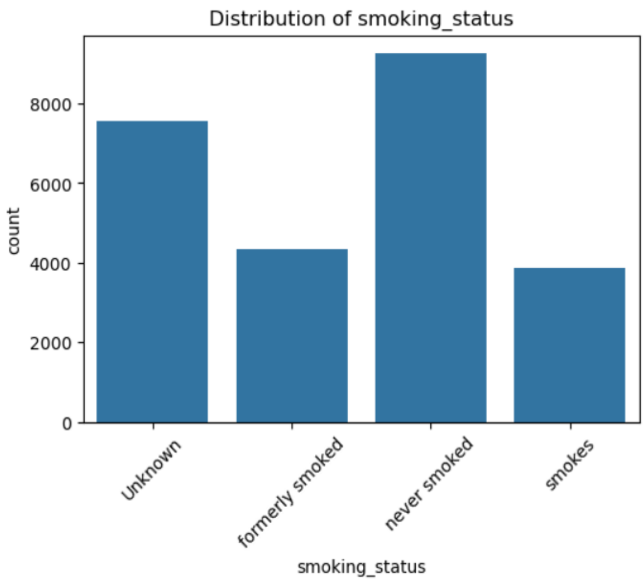

Figure19- Distribution of smoking status

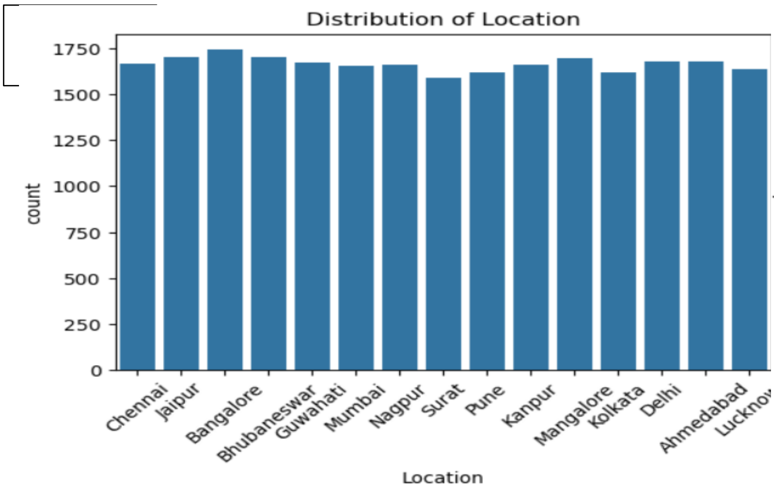

Figure20- Distribution of location

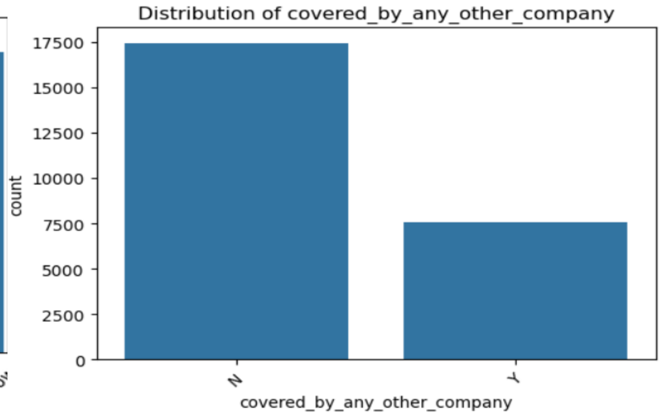

Figure21- Distribution of covered by any other company

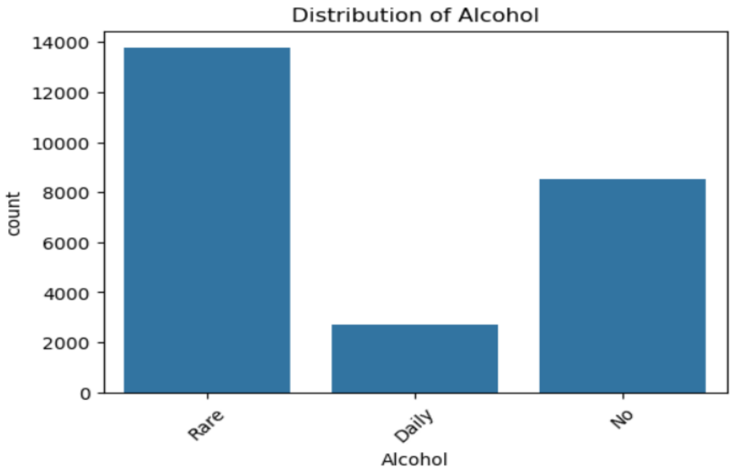

Figure22- Distribution of alcohol

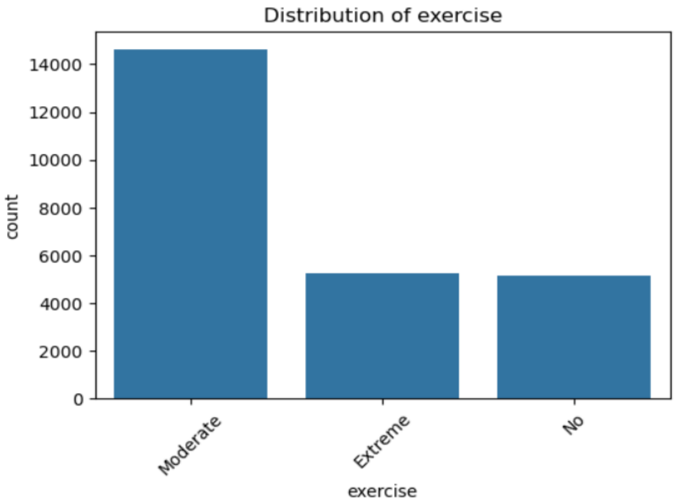

Figure23- Distribution of exercise

>Observations;

- Most of the customer are not involve in adventure sports.
- Occupationally students are more than business and salried.
- Most of the customer have no heart decs history.
- Most of the customer have no other major decs history.
- Male customers are more than female customers.
- Never smoked customer are higher than unknown, formerly smoke and smoked, smoked persons are very compared to other lower than 400.
- Approximately all location had same number of hospitals.
- Most of the customer had not covered by any other insurance company. Bellow 7500 customers are covered by other insurance company.
- Daily alcohol consumption customers are very lower than the never and rare, rare alcohol consumption customers are higher.
- Customers who are moderate in exercise are very higher than extreme and never do exercise customers.

b) Bivariate Analysis

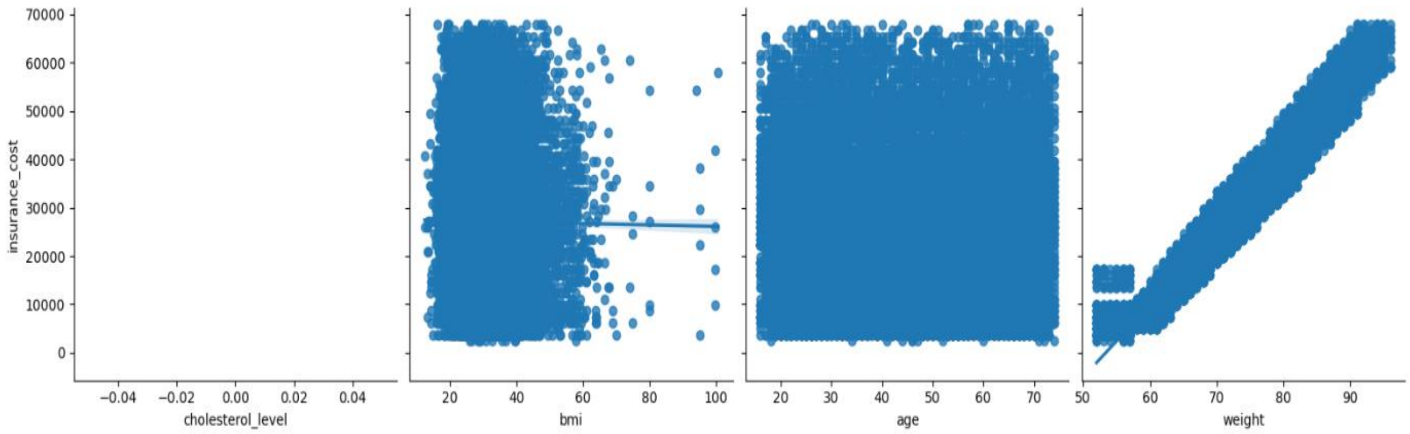

Figure24- Insurance cost vs cholesterol level, bmi, age, weight

>Observations;

- The scatter plot with regression for BMI likely shows a positive correlation as well.
- A clear upward trend would indicate that as BMI increases, insurance costs tend to rise.
- Age is often positively correlated with insurance cost due to the increased risk of health issues with aging.
- A strong positive trend here would suggest that older individuals tend to have higher insurance costs.
- The trend in the weight plot will help indicate there is a positive correlation between weight and insurance costs.
- A positive correlation would imply that higher weights might be associated with higher insurance costs.

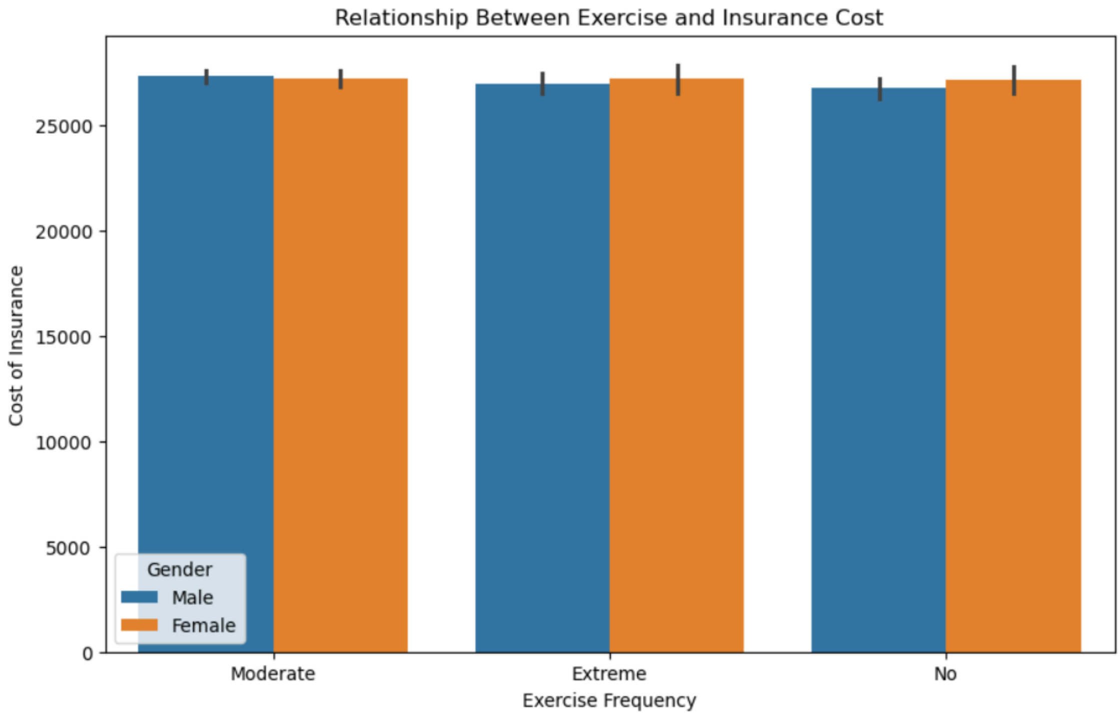

Figure25- Relationship Between Exercise and Insurance Cost

>Observations; Female customers who have never do exercise they have more chance for health risk and more chance to take insurance.

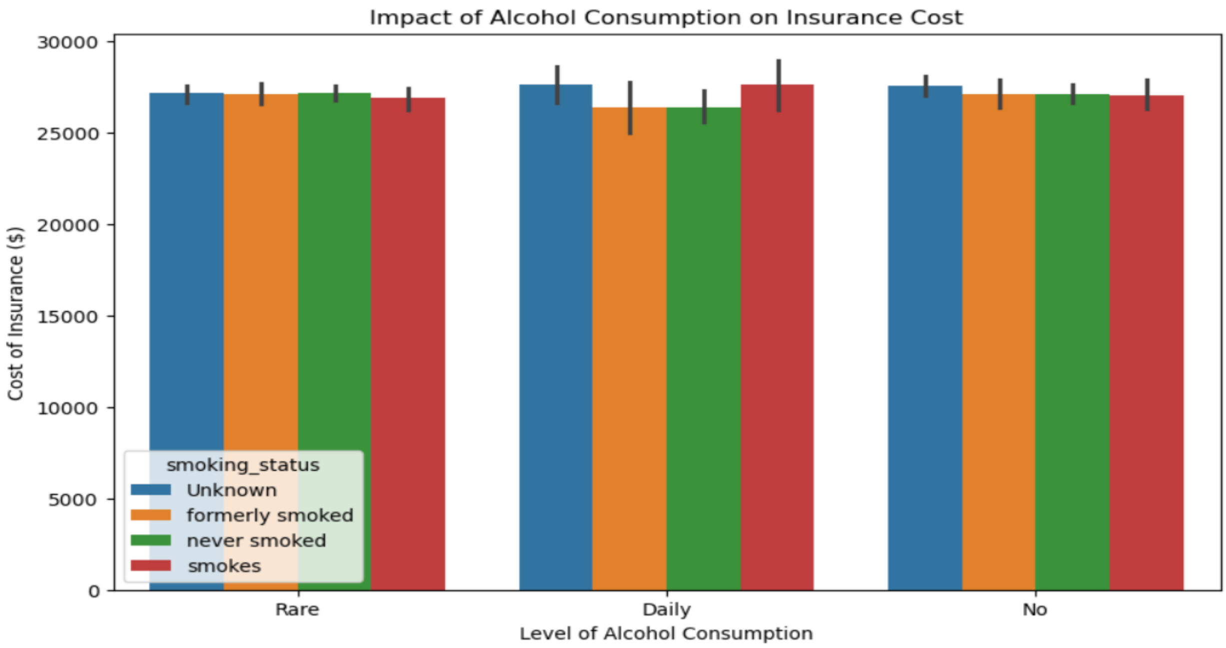

Figure26- Impact of Alcohol Consumption on Insurance Cost

>Observations; Customers who had consuming alcohol daily and also smoke has the chance of health risk more they use more insurance policies.

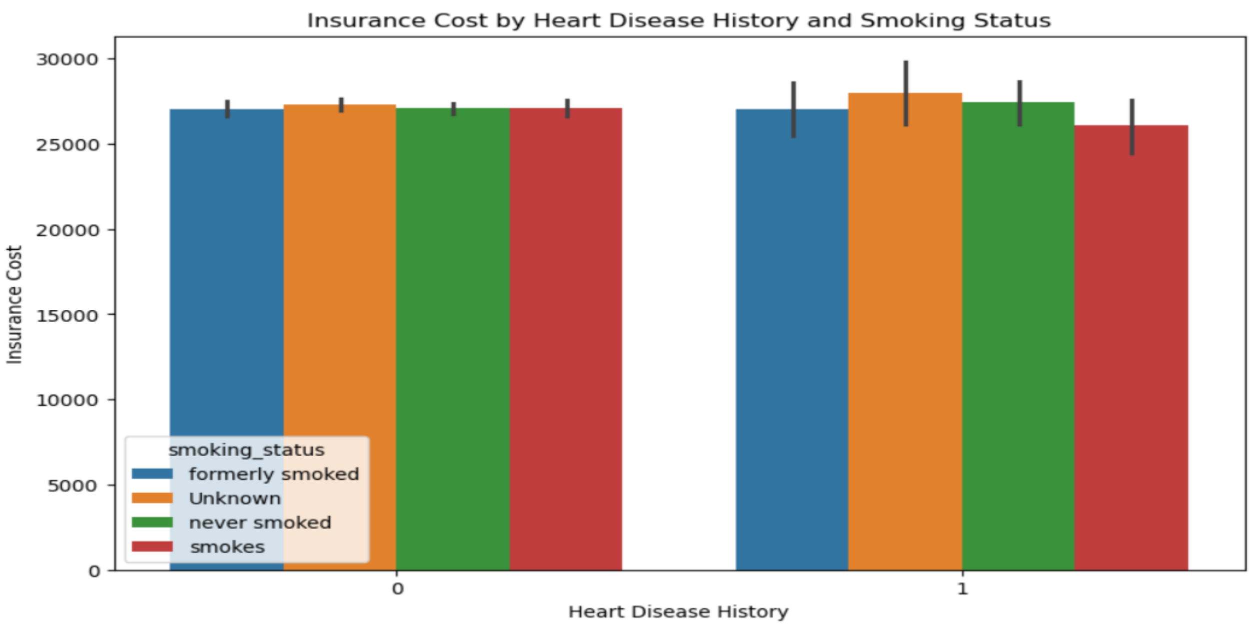

Figure27- Insurance cost by heart disease and smoking status

>Observations; Customers with no heart disease history but smoked is lower than customers who have no heart disease history but smoked is higher.

4. Business insights from EDA

a) Is the data unbalanced? If so, what can be done? Please explain in the context of the business b) Any business insights using clustering (if applicable) c) Any other business insights

-Check for data imbalance

```
Value Counts for Categorical Variables:
Gender
Male      16422
Female     8578
Name: count, dtype: int64
smoking_status
never smoked       9249
Unknown            7555
formerly smoked    4329
smokes             3867
Name: count, dtype: int64
Location
Bangalore      1742
Jaipur         1706
Bhubaneswar    1704
Mangalore      1697
Delhi          1680
Ahmedabad      1677
Guwahati       1672
Chennai        1669
Kanpur         1664
Nagpur         1663
Mumbai         1658
Lucknow        1637
Pune           1622
Kolkata        1620
Surat          1589
Name: count, dtype: int64
```

>Observations;

* Gender:
  - There are nearly twice as many males (16,422) as females (8,578) in the dataset, which may affect gender-based analyses.

* Smoking Status:
  - Most customers around 9,249 have never smoked.
  - A large portion around 7,555 has an "Unknown" smoking status, which could complicate analysis.
  - There's a similar count for those who currently smoke around 3,867 and those who formerly smoked around 4,329.

* Location:
  - The data is spread across multiple cities, with Bangalore having the most entries around 1,742 and Surat the fewest nearly 1,589.

-Clustering on numerical variables only

- Features like age, BMI, and insurance cost have different ranges, which can affect clustering results. Without scaling, features with larger ranges could dominate the clustering process, giving skewed results.
- Using StandardScaler changes each feature to have a mean of 0 and a standard deviation of 1. This puts all features on the same scale, making them easier to compare directly.
- After scaling, the data values are now in standard deviations from the mean.
- We are using KMeans clustering for these three numerical clusters ('age', 'bmi', 'insurance_cost')

- Visualizing clusters

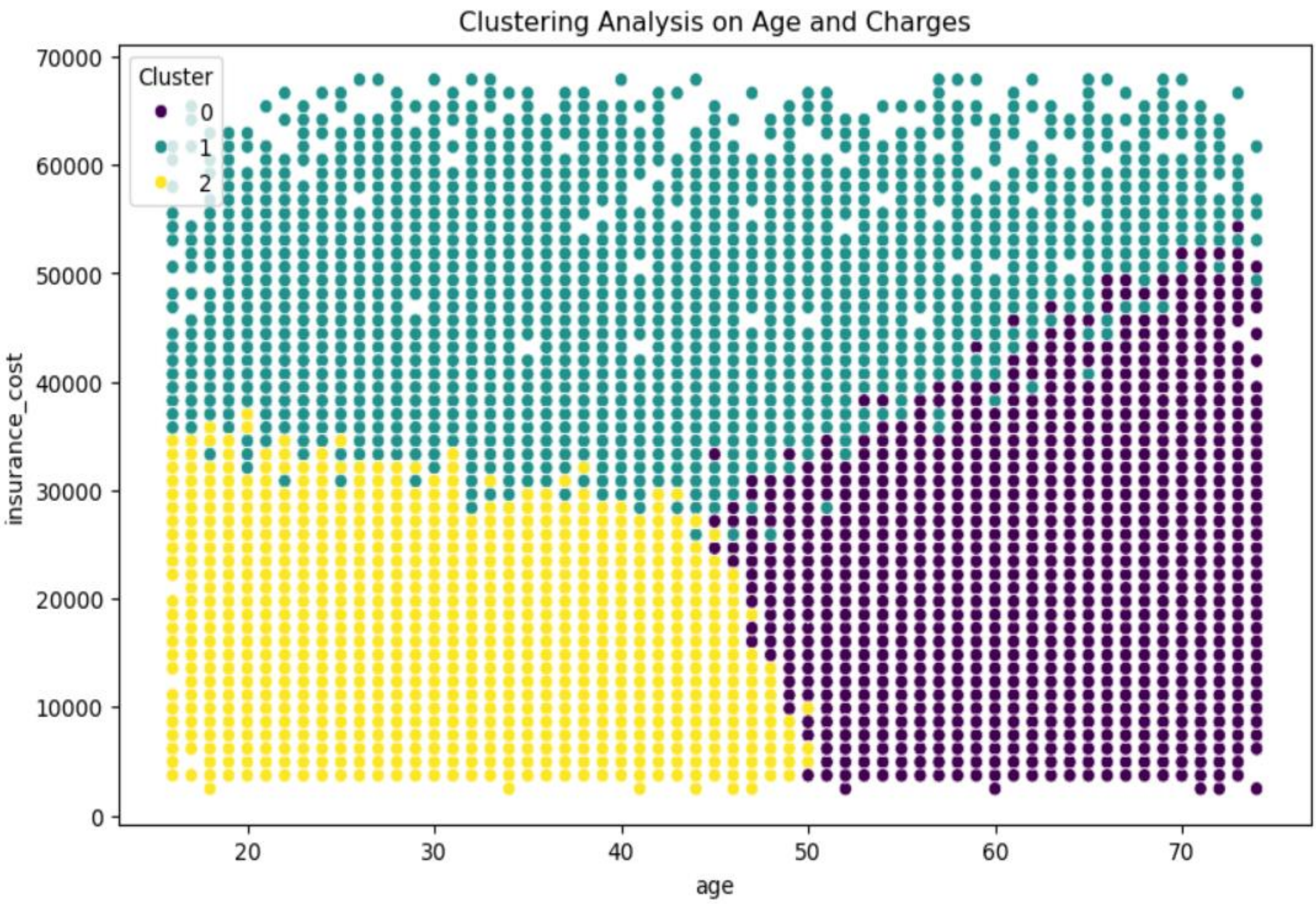

Figure28- Clustering Analysis on Age and Insurance cost

>Observations; This scatter plot shows a clustering of people based on their age and insurance costs, divided into three groups which is called clusters and it is also shown in different colors.

Cluster 0 which color is Purple:

- Mostly older customers around age 50 and above 50.
- It shows that they have a wide range of insurance costs, often in the mid to high range.

Cluster 1 which color is Teal:

- It includes customers of all ages.
- Most of the customer have medium to high insurance costs.
- This cluster 1 may have higher insurance needs, possibly due to lifestyle or health issues.

Cluster 2 which color is Yellow:

- Mainly younger customer under 40.
- It shows that they have lower insurance costs.
- Likely represents younger people with fewer health needs.

Overall Patterns from those clusters:

- Older customers are mostly in the clusters with higher insurance costs that is clusters 0 and 1.
- Younger customers are generally in the low-cost cluster that is cluster 2.

-Further insights may involve analyzing the differences in charges between smokers and non-smokers, as well as identifying regions associated with higher charges

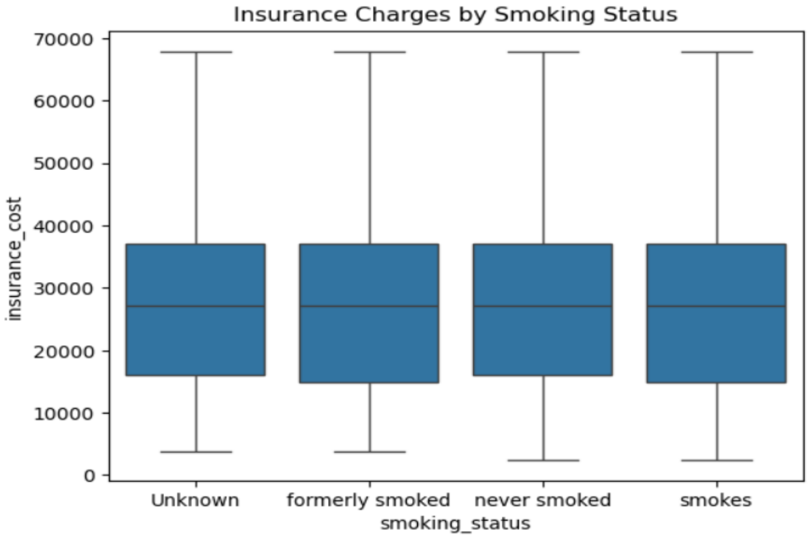

Figure29- Insurance charges by smoking status

>Observations;

- Insurance costs are similar for all smoking statuses like Unknown, formerly smoked, never smoked, and smokes.
- The box plots show there is a big range in insurance costs within each group.
- Smoking status doesn't seem to have a big impact on insurance costs in this data.

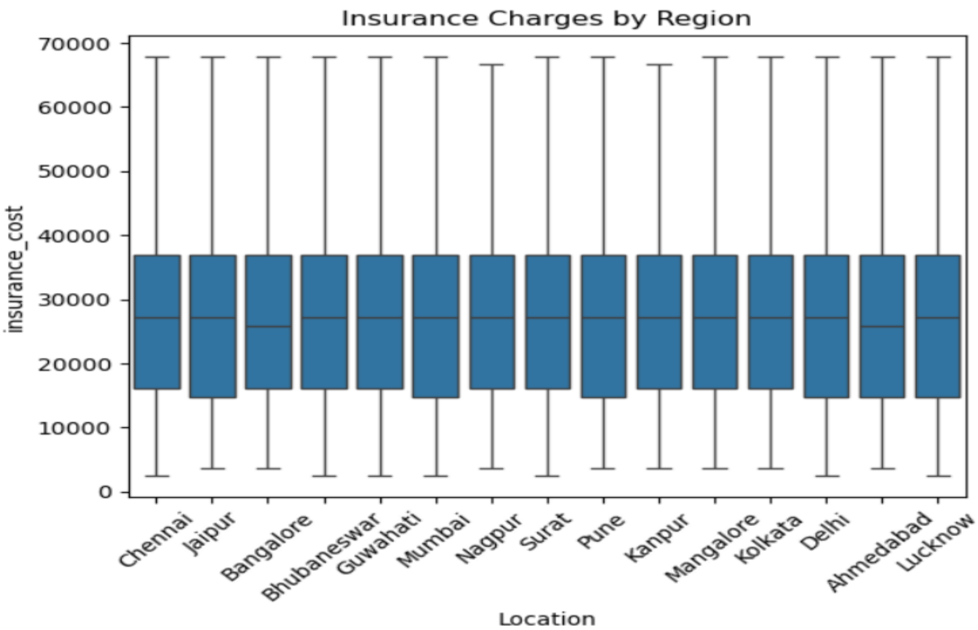

Figure30- Insurance charges by region

>Observations;

- Insurance costs appear similar across all locations, with no region showing notably higher or lower costs.
- Every region has a broad range of insurance costs, as shown by the long whiskers.
- Location doesn’t seem to have a major impact on insurance costs in this data.